We will come back at 10 a.m.

Lecture 22: Three-Dimensional Coordinate Systems

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2024/2025

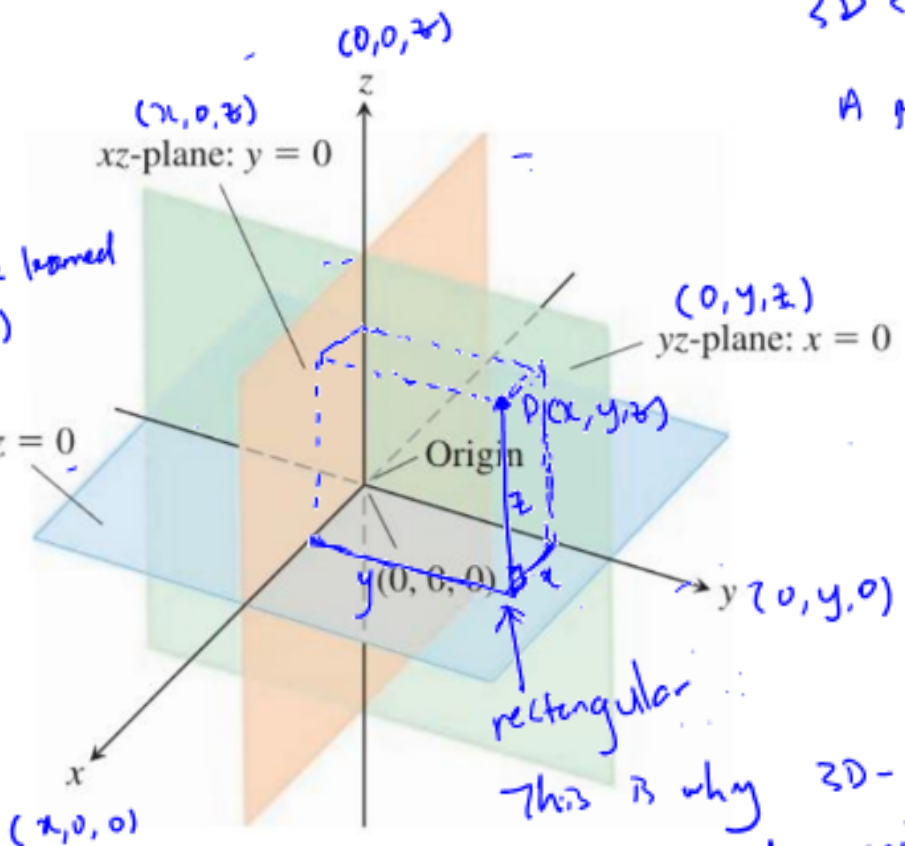
Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define and identify the three-dimensional Cartesian coordinate system.
- ☐ Locate points in space using ordered triples (x, y, z).
- ☐ Describe the coordinate axes, coordinate planes, and the concept of octants.
- ☐ Determine the distance between two points in space.
- ☐ Recognize the equation of a sphere given its center and radius.

3-Dimensional Coordinates System

Previously — 2D



Here is what we learned before (x, y, 0)

3D space.

A point in 3D-space is written as

P(x, y, z)

x-component

y-component

z-component.

z - perpendicular distance to xy-plane

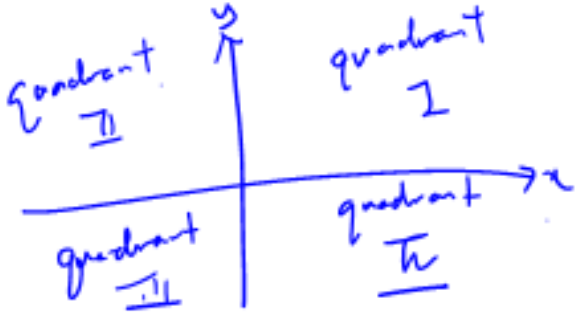
y- perpendicular distance to xz-plane

x- perpendicular distance to yz-plane.

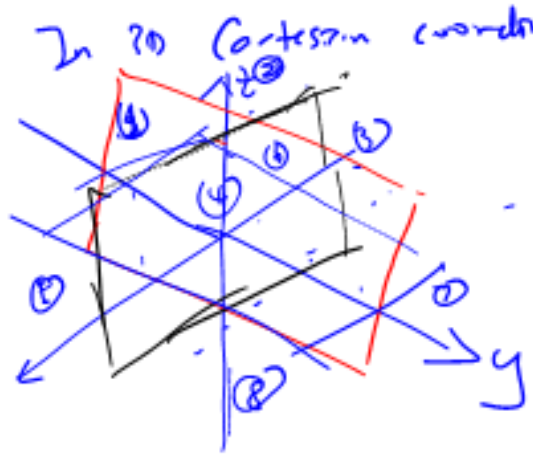
rectangular

This is why 3D-cartesian coordinate for space is also called rectangular coordinate.

In 2D Cartesian coordinate



In 3D Cartesian coordinate



First octant. 8 octants

→ Octant I : $\{(x, y, z) : x \geq 0, y \geq 0, z \geq 0\}$

$\{(x, y, z) : x \geq 0, y \geq 0, z \leq 0\}$

$\{(x, y, z) : x \geq 0, y \leq 0, z \geq 0\}$

$\{(x, y, z) : x \leq 0, y \geq 0, z \geq 0\}$

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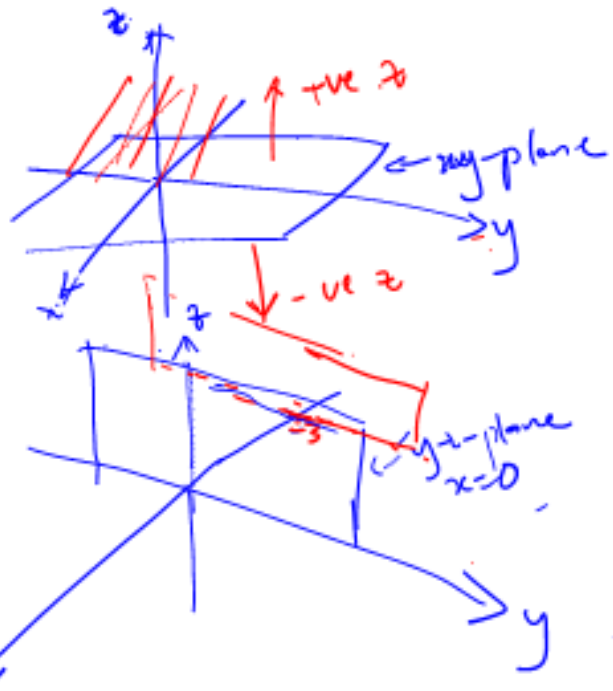
$\{(x, y, z) : x \leq 0, y \leq 0, z \leq 0\}$

3-Dimensional Coordinates

Example: Give the geometric interpretation of the following equations and inequalities.

(a) $z \geq 0$

$z=0$ ⇒ our xy-plane



All the points on and above the xy-plane.

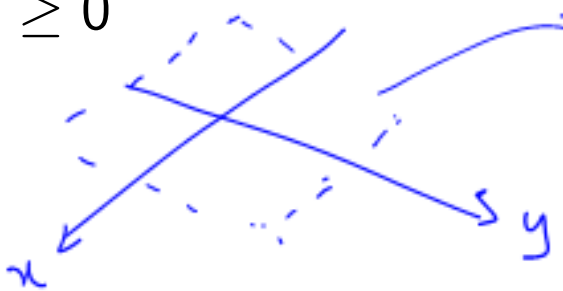
(b) $x = -3$

$x=0$ ⇒ our yz-plane.

All the points on a plane parallel to and 3 units behind the yz-plane (perpendicular to the line $x = -3$)

(c) $z = 0, x \leq 0, y \geq 0$

xy plane





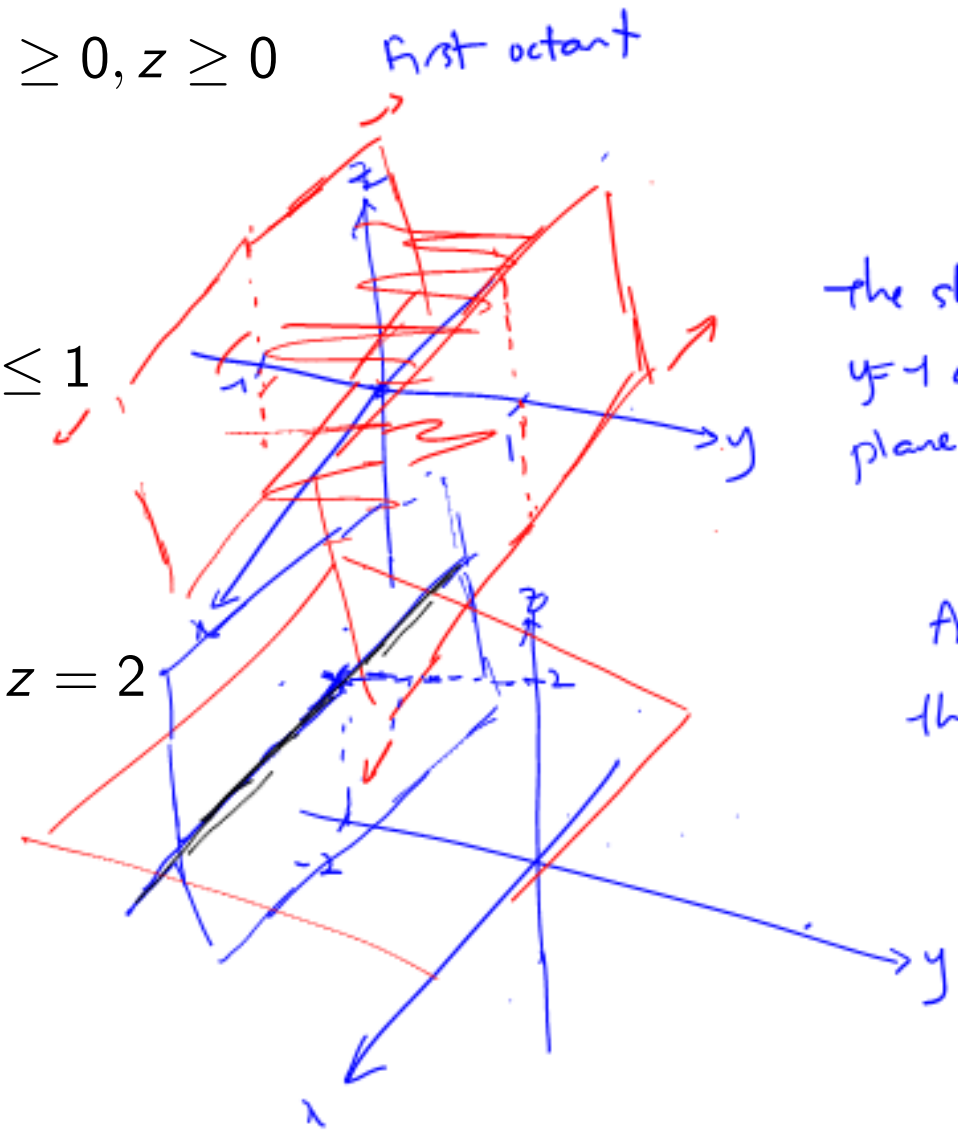
Second quadrant of xy-plane.

3-Dimensional Coordinates

(d) $x \geq 0, y \geq 0, z \geq 0$

(e) $-1 \leq y \leq 1$

(f) $y = -2, z = 2$



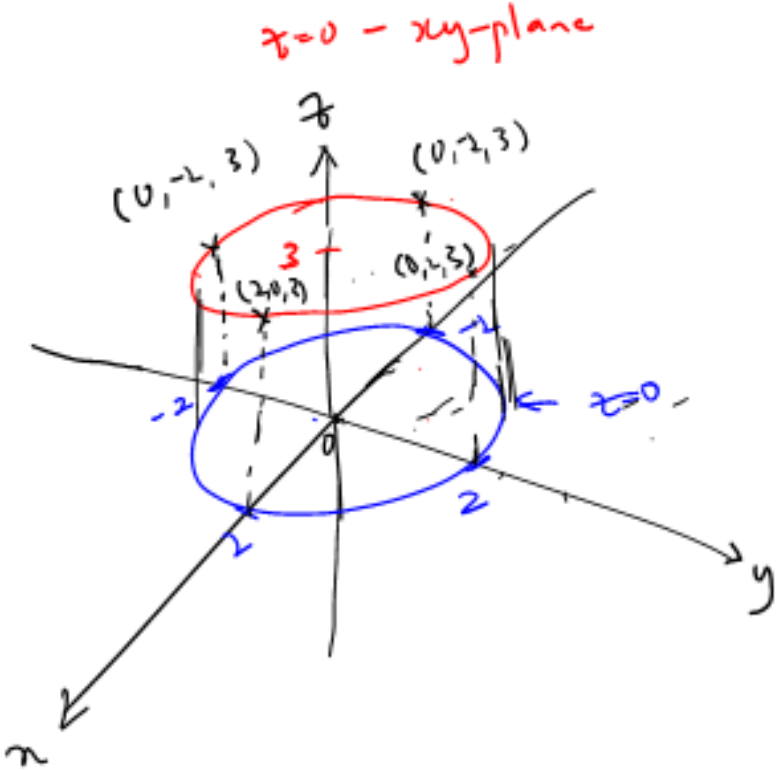
the slab between the planes $y=-1$ and $y=1$, including the planes.

A line of intersection of the planes $z=2$ and $y=-2$

3-Dimensional Coordinates

Example: What points (x, y, z) satisfy the equations

$$x^2 + y^2 = 4 \quad \text{and} \quad z = 3?$$



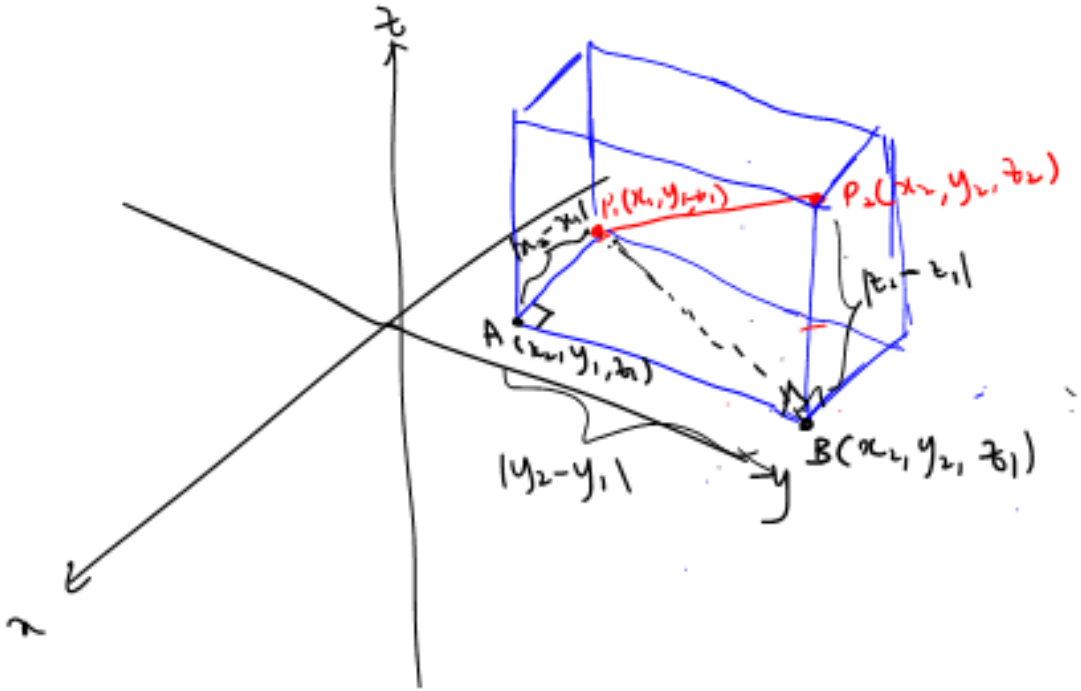
The circle $x^2+y^2=4$ in the plane $z=3$.

$0 \le z \le 3$
This will form a cylinder.

Distance Between Points in Space

The Distance Between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



$$|P_1P_2|^2 = |P_1B|^2 + |P_2B|^2$$

$$= |P_1B|^2 + |z_2 - z_1|^2$$

$$= |P_1A|^2 + |AB|^2 + |z_2 - z_1|^2$$

$$= |x_2 - x_1|^2 + |y_2 - y_1|^2 + |z_2 - z_1|^2$$

$$= (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$$

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Distance Between Points in Space

Example: Find the distance between $P_1(2, 1, 5)$ and $P_2(-2, 3, 0)$.

$$|P_1P_2| = \sqrt{(-2-2)^2 + (3-1)^2 + (0-5)^2}$$

$$= \sqrt{16 + 4 + 25}$$

$$= \sqrt{45} = 3\sqrt{5}$$

Spheres

In xy-plane (2D) the collection of points with the same distance from a particular point forms a circle.

The collection of points with the same distance from a particular point forms a sphere.

distance → radius. particular point → center

Standard Equation for the Sphere of Radius a and Center (x_0, y_0, z_0)

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

derived from the equation for distance between two points.

Spheres

Example: Find the center and radius of the sphere

$$x^2 + y^2 + z^2 + 3x - 4z + 1 = 0.$$

$x^2 + 3x + y^2 + z^2 - 4z + 1 = 0$

$\left(x + \frac{3}{2}\right)^2 - \frac{9}{4} + y^2 + (z-2)^2 - 4 + 1 = 0$

$\left(x + \frac{3}{2}\right)^2 + y^2 + (z-2)^2 = \frac{21}{4}$

Sphere with radius $\sqrt{\frac{21}{4}}$ centered at $\left(\frac{-3}{2}, 0, 2\right)$

Spheres

Example: Give the geometric interpretation of the following equations and inequalities.

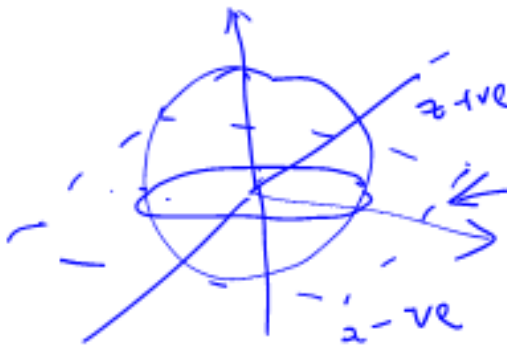
$x^2 + y^2 + z^2 = 4$ → a sphere with radius 2 centered at origin.

(a) $x^2 + y^2 + z^2 < 4$ interior of the sphere (not including the sphere).

(b) $x^2 + y^2 + z^2 \leq 4$ the sphere including its interior

(c) $x^2 + y^2 + z^2 > 4$ exterior of the sphere (not including the sphere)

(d) $x^2 + y^2 + z^2 = 4,\ z \leq 0$



lower hemisphere cut by the xy-plane.